

For **15th practical (Restore Database)**, you FIRST need a backup file.

Without backup file, restore cannot happen.

So actual flow is:

REAL FLOW OF RESTORE PRACTICAL

Create Database

- Create Table
- Insert Records
- Take Backup
- Drop Database
- Restore Database
- Check Restored Data

That is the proper logic.

SIMPLE COMPLETE FLOW OF 15

STEP 1 — Open CMD

cmd

STEP 2 — Go to MySQL bin Folder

```
cd C:\Program Files\MySQL\MySQL Server 8.0\bin
```

STEP 3 — Open MySQL

```
mysql -u system -p
```

Password:

```
root
```

STEP 4 — Create Database

```
CREATE DATABASE collegeDB;
```

STEP 5 — Use Database

```
USE collegeDB;
```

STEP 6 — Create Table

```
CREATE TABLE Student(  
    StudentID INT PRIMARY KEY,  
    Name VARCHAR(50)  
);
```

STEP 7 — Insert Records

```
INSERT INTO Student VALUES  
(1, 'Tanmay'),  
(2, 'Rahul');
```

STEP 8 — Display Table

```
SELECT * FROM Student;
```

You will see records.

STEP 9 — Exit MySQL

```
EXIT;
```

STEP 10 — Take Backup

```
mysqldump -u system -p collegeDB > college_backup.sql
```

Password:

root

Now backup file is created.

STEP 11 — Open MySQL Again

```
mysql -u system -p
```

Password:

root

STEP 12 — Drop Database

```
DROP DATABASE collegeDB;
```

Now database deleted.

STEP 13 — Create Empty Database Again

```
CREATE DATABASE collegeDB;
```

STEP 14 — Exit MySQL

```
EXIT;
```

STEP 15 — Restore Database

```
mysql -u system -p collegeDB < college_backup.sql
```

Password:

root

Now backup restored.

STEP 16 — Open MySQL Again

```
mysql -u system -p
```

Password:

```
root
```

STEP 17 — Use Database

```
USE collegeDB;
```

STEP 18 — Check Restored Table

```
SELECT * FROM Student;
```

You will again see:

- 1 Tanmay
- 2 Rahul

That proves restore successful.

WHY DROP DATABASE?

External may ask this.

Answer:

“To check whether backup can recover deleted database data.”

Very good viva answer.

11. Database Backup — Complete Practical

Question

Write the command to take backup of MySQL database collegeDB using mysqldump.

Real Flow of Practical

Open CMD

→ Open MySQL

- Create Database
- Use Database
- Create Table
- Insert Records
- Display Table
- Exit MySQL
- Run mysqldump command
- Backup file created

Complete Demonstration

STEP 1 — Open CMD

Press:

Windows + R

Type:

cmd

Press Enter.

STEP 2 — Go to MySQL bin Folder

cd C:\Program Files\MySQL\MySQL Server 8.0\bin

STEP 3 — Open MySQL

```
mysql -u system -p
```

Press Enter.

STEP 4 — Enter Password

```
root
```

STEP 5 — Create Database

```
CREATE DATABASE collegeDB;
```

STEP 6 — Use Database

```
USE collegeDB;
```

STEP 7 — Create Table

```
CREATE TABLE Student(  
    StudentID INT PRIMARY KEY,  
    Name VARCHAR(50)  
);
```

STEP 8 — Insert Records

```
INSERT INTO Student VALUES  
(1, 'Tanmay'),  
(2, 'Rahul');
```

STEP 9 — Display Table

```
SELECT * FROM Student;
```

Output:

```
+-----+-----+  
| StudentID | Name |  
+-----+-----+  
| 1         | Tanmay |
```

STEP 10 — Exit MySQL

EXIT;

Now you are back in CMD.

STEP 11 — Take Backup

```
mysqldump -u system -p collegeDB > college_backup.sql
```

Press Enter.

STEP 12 — Enter Password

root

STEP 13 — Backup File Created

File created:

college_backup.sql

inside current folder.

STEP 14 — Verify Backup File (Optional)

You can type:

dir

You will see:

college_backup.sql

That confirms backup successful.

Viva Questions and Answers

Q1. What is mysqldump?

Answer:

mysqldump is a utility used to take backup of MySQL database.

Q2. Where is mysqldump executed?

Answer:

In CMD or terminal, not inside MySQL.

Q3. What is created after backup?

Answer:

A .sql backup file is created.

Q4. How do you know backup succeeded?

Answer:

Backup .sql file gets created in current folder.

MOST IMPORTANT THING

This command:

```
mysqldump -u system -p collegeDB > college_backup.sql
```

must be run in CMD.

NOT inside:

```
mysql>
```

14. Backup studentDB — Complete Practical

Question

Write the command to take backup of MySQL database studentDB using mysqldump.

Real Flow of Practical

Open CMD

- Open MySQL
 - Create Database
 - Use Database
 - Create Table
 - Insert Records
 - Display Table
 - Exit MySQL
 - Run mysqldump command
 - Backup file created
-

Complete Demonstration

STEP 1 — Open CMD

Press:

Windows + R

Type:

cmd

Press Enter.

STEP 2 — Go to MySQL bin Folder

```
cd C:\Program Files\MySQL\MySQL Server 8.0\bin
```

STEP 3 — Open MySQL

```
mysql -u system -p
```

Press Enter.

STEP 4 — Enter Password

root

STEP 5 — Create Database

```
CREATE DATABASE studentDB;
```

STEP 6 — Use Database

```
USE studentDB;
```

STEP 7 — Create Table

```
CREATE TABLE Student(  
    RollNo INT PRIMARY KEY,  
    Name VARCHAR(50)  
);
```

STEP 8 — Insert Records


```
INSERT INTO Student VALUES
(1,'Tanmay'),
(2,'Rahul');
```

STEP 9 — Display Table

```
SELECT * FROM Student;
```

Output:

```
+-----+-----+
| RollNo | Name  |
+-----+-----+
| 1      | Tanmay |
| 2      | Rahul  |
+-----+-----+
```

STEP 10 — Exit MySQL

```
EXIT;
```

Now you are back in CMD.

STEP 11 — Take Backup

```
mysqldump -u system -p studentDB > student_backup.sql
```

Press Enter.

STEP 12 — Enter Password

root

STEP 13 — Backup File Created

File created:

student_backup.sql

inside current folder.

STEP 14 — Verify Backup File (Optional)

Type:

dir

You will see:

`student_backup.sql`

That confirms backup successful.

Viva Questions and Answers

Q1. What is mysqldump?

Answer:

mysqldump is used to take backup of MySQL database.

Q2. Where is mysqldump executed?

Answer:

In CMD or terminal.

Q3. What file is created after backup?

Answer:

A .sql backup file.

Q4. Why do we take backup?

Answer:

To recover data if database gets deleted or corrupted.

IMPORTANT

This command:

```
mysqldump -u system -p studentDB > student_backup.sql
```

must be executed in CMD.

NOT inside:

```
mysql>
```

YES. Exactly.

If your MySQL application already opens directly into CMD-like MySQL terminal, then you
MOST LIKELY do NOT need:

```
cd C:\Program Files\MySQL...
```

That step is only for people using normal CMD separately.

Your flow is simpler.

YOUR ACTUAL FLOW

STEP 1

Open MySQL application.

It asks:

Enter username:

You type:

system

Then password:

root

STEP 2

Now you are inside MySQL:

mysql>

Now perform queries:

- CREATE DATABASE
 - USE
 - CREATE TABLE
 - INSERT
 - SELECT
-

STEP 3

Then type:

EXIT;

Now you return to normal CMD.

STEP 4

NOW run backup command directly:

For 11

```
mysqldump -u system -p collegeDB > college_backup.sql
```

OR

For 14

```
mysqldump -u system -p studentDB > student_backup.sql
```

STEP 5

Enter password:

root

STEP 6

Now type:

dir

You will see:

college_backup.sql

or

student_backup.sql

That means DONE.

IMPORTANT CORRECTION

You said:

“after restoring I type dir”

Actually:

dir is mainly for checking BACKUP FILE

Not restore.

RESTORE FLOW DIFFERENT

For restore:

1. Exit MySQL
2. Run:

```
mysql -u system -p collegeDB < college_backup.sql
```

3. Enter password
4. Open MySQL again
5. Run:

USE collegeDB;

SELECT * FROM Student;

to verify restored data.

FINAL SIMPLE UNDERSTANDING

Task	After EXIT do what?
Backup	Run mysqldump
Check backup file	Run dir
Restore	Run mysql < backup.sql
Check restored data	Open MySQL again and use SELECT *